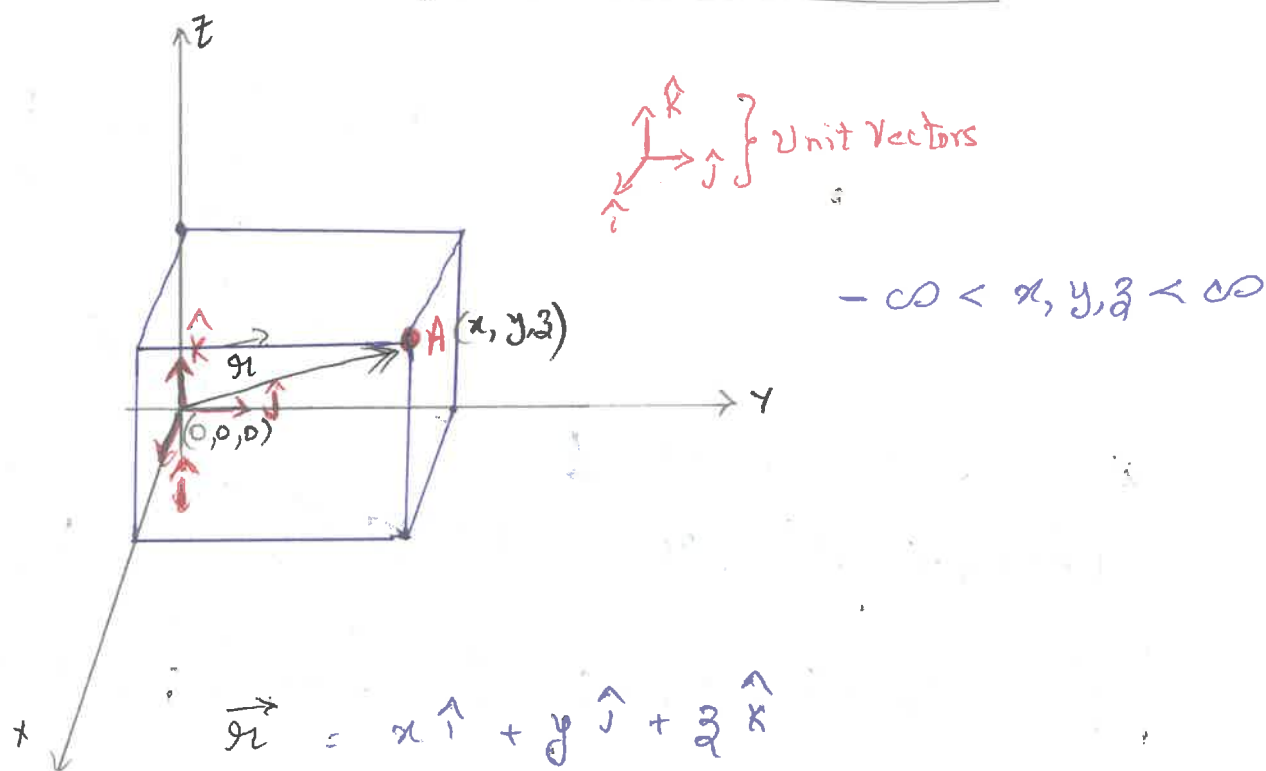


Coordinate Systems: Alternate notation.

(1)



$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

↳ Position vector

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

$$\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$$

$$\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$

$$\vec{F} = m\vec{a} = ma_x\hat{i} + ma_y\hat{j} + ma_z\hat{k}$$

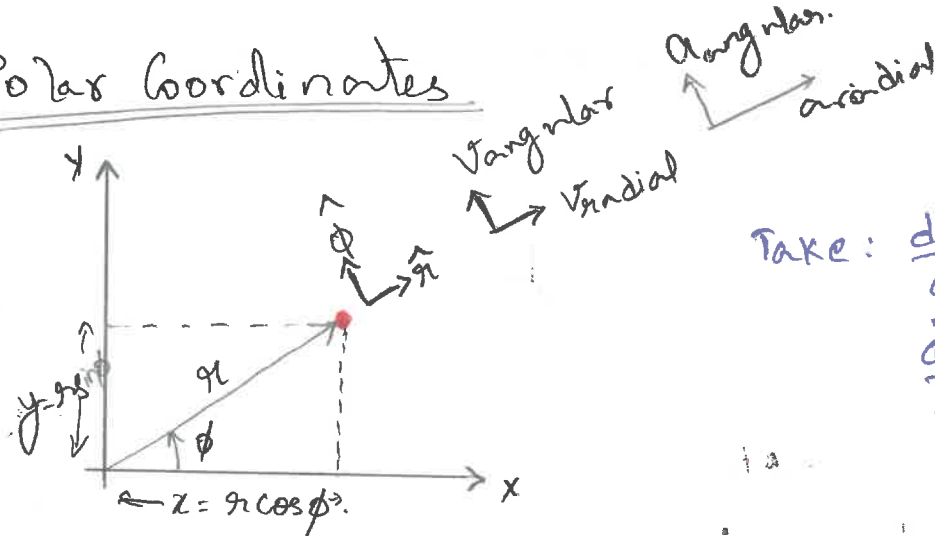
$$\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$$

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} \text{ } \{ \text{line element.} \}$$

$$\text{displacement vector } \{ \vec{r}_f - \vec{r}_i = (x_f - x_i)\hat{i} + (y_f - y_i)\hat{j} + (z_f - z_i)\hat{k} \}$$

$\vec{r}_f \rightarrow$ final position vector
 $\vec{r}_i =$ initial position vector

Polar Coordinates



Take: $\frac{dr}{dt} = v_r$ $\frac{d\hat{r}}{dt} = a_r$

$\frac{d\phi}{dt} = \omega$

$\frac{d\omega}{dt} = \alpha$

$0 \leq r < \infty$
 $0 \leq \phi < 2\pi$
 ↳ range of r & ϕ

$x = r \cos \phi$ $y = r \sin \phi$

$|\vec{r}| = \sqrt{x^2 + y^2} = r$

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

$\hat{r} = \cos \phi \hat{i} + \sin \phi \hat{j}$ $\hat{\phi} = -\sin \phi \hat{i} + \cos \phi \hat{j}$

$\vec{r} = r \hat{r}$ } Position vector in polar coordinates

$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d(r \hat{r})}{dt} = \frac{dr}{dt} \hat{r} + r \frac{d\hat{r}}{dt}$

$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt} \hat{r} + r \frac{d(\cos \phi \hat{i} + \sin \phi \hat{j})}{dt}$

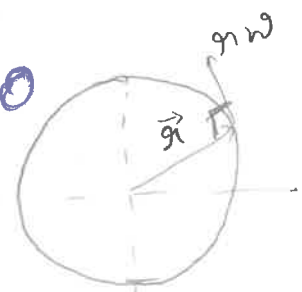
Now $\frac{dr}{dt} = v_r$

$\therefore \vec{v} = v_r \hat{r} + r (-\sin \phi \hat{i} + \cos \phi \hat{j}) \frac{d\phi}{dt}$

$\vec{v} = v_r \hat{r} + r \omega \hat{\phi}$

Now if r is constant $\Rightarrow v_r = \frac{dr}{dt} = 0$

$\vec{v} = r \omega \hat{\phi} \Rightarrow$ The body is in uniform circular motion



→ To find acceleration in polar coordinates (3)

$$\vec{v} = v_r \hat{r} + r \omega \hat{\phi} \quad \leftarrow \text{do a time derivative of velocity.}$$

$$\frac{d\vec{v}}{dt} = \frac{d(v_r \hat{r})}{dt} + \frac{d(r \omega \hat{\phi})}{dt}$$

$$\frac{d\vec{v}}{dt} = \vec{a} \quad \frac{dv_r}{dt} = a_r \quad \frac{d\omega}{dt} = \alpha \quad \left\{ \text{using these} \right.$$

$$\frac{d\vec{v}}{dt} = \frac{dv_r}{dt} \hat{r} + v_r \frac{d\hat{r}}{dt} + \frac{dr}{dt} \omega \hat{\phi} + r \frac{d\omega}{dt} \hat{\phi} + r \omega \frac{d\hat{\phi}}{dt}$$

$$\vec{a} = a_r \hat{r} + v_r \omega \hat{\phi} + v_r \omega \hat{\phi} + r \alpha \hat{\phi} + r \omega \frac{d\hat{\phi}}{dt}$$

$$\vec{a} = a_r \hat{r} + 2 v_r \omega \hat{\phi} + r \alpha \hat{\phi} + r \omega (-\omega \hat{r})$$

$$\vec{a} = (a_r - r \omega^2) \hat{r} + (2 v_r \omega + r \alpha) \hat{\phi}$$

radial component

angular component

$2 v_r \omega$ is Coriolis acceleration.

→ If $\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$

$$\vec{\omega} = \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}$$

Coriolis accn is given as $2(\vec{v} \times \vec{\omega}) = 2$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_x & v_y & v_z \\ \omega_x & \omega_y & \omega_z \end{vmatrix}$$

For Cylindrical Coordinates

$$\vec{r} = r \hat{r} + z \hat{z} \rightarrow \text{position vector}$$

$$\vec{v} = v_r \hat{r} + r\omega \hat{\phi} + v_z \hat{z} \rightarrow \text{velocity}$$

$$\vec{a} = (a_r - r\omega^2) \hat{r} + (r\alpha + 2v_r\omega) \hat{\phi} + a_z \hat{z} \rightarrow \text{acceleration}$$

We just add z component to the planar Polar Coordinates to extend into cylindrical Coordinates.

Spherical Coordinates

$$\vec{r} = r \hat{r} \rightarrow \text{position vector}$$

$$\hat{r} = \sin\theta \cos\phi \hat{i} + \sin\theta \sin\phi \hat{j} + \cos\theta \hat{k}$$

$$\hat{\theta} = \cos\theta \cos\phi \hat{i} + \cos\theta \sin\phi \hat{j} - \sin\theta \hat{k}$$

$$\hat{\phi} = -\sin\phi \hat{i} + \cos\phi \hat{j}$$

Three unit vectors.

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt} \hat{r}$$

$$\text{Take } \begin{cases} \frac{dr}{dt} = v_r \\ \frac{d\phi}{dt} = \omega \\ \frac{d\theta}{dt} = v_\theta \end{cases}$$

$$\frac{d\hat{r}}{dt} = \frac{d}{dt} \{ \sin\theta \cos\phi \hat{i} + \sin\theta \sin\phi \hat{j} + \cos\theta \hat{k} \}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d(r\hat{r})}{dt} = \frac{dr}{dt} \hat{r} + r \frac{d\hat{r}}{dt}$$

$$\boxed{\vec{v} = v_r \hat{r} + r v_\theta \hat{\theta} + r \sin\theta \omega \hat{\phi}}$$

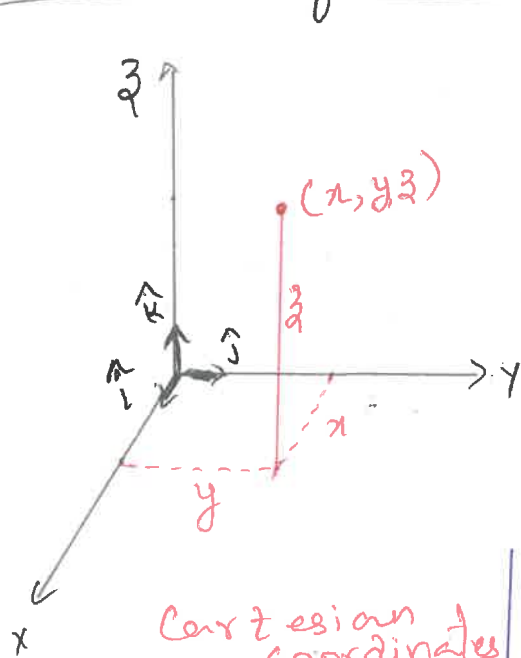
radial component

θ component

ϕ component.

Conversion of 3D-Coordinates

5



Cartesian coordinates

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

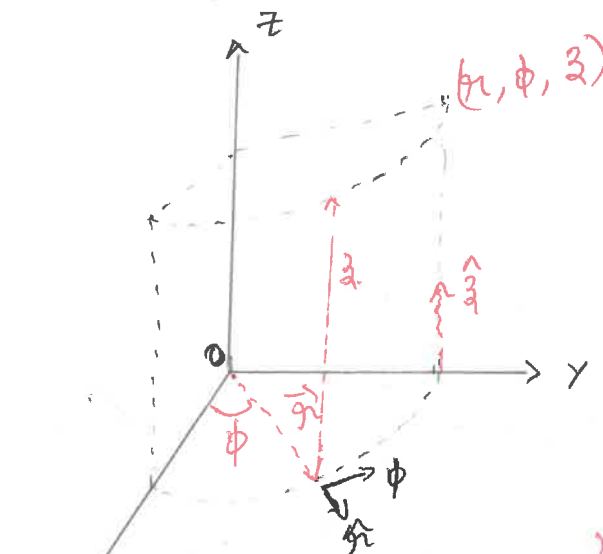
$$-\infty < x < +\infty$$

$$-\infty < y < +\infty$$

$$-\infty < z < +\infty$$

$$\hat{i}, \hat{j}, \hat{k}$$

(Unit vectors)



Cylindrical coordinates

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$z = z$$

$$0 \leq r < \infty$$

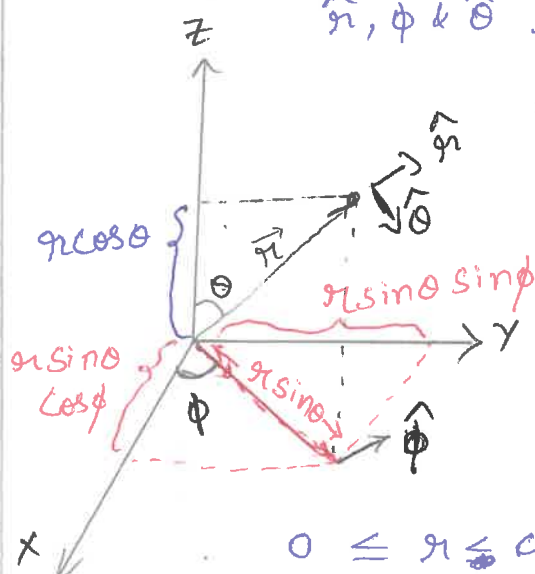
$$0 \leq \phi \leq 2\pi$$

$$-\infty < z < +\infty$$

$$\hat{r}, \hat{\phi}, \hat{z}$$

(Unit vectors)

$\hat{r}, \hat{\phi}, \hat{\theta}$ } unit vectors



$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\tan \phi = \frac{y}{x} \Rightarrow \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = \cos^{-1}\left\{\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right\}$$

Cartesian & spherical coordinates conversion

$$0 \leq r < \infty$$

$$0 \leq \phi \leq 2\pi$$

$$0 \leq \theta \leq \pi$$

1) If (i) $(x, y) = (2, 4)$ (ii) $(x, y) = (3, 6)$ & (iii) $(x, y) = (-4, -2)$. What are the polar coordinates & their unit vectors?

Sol (i) $(x, y) = (2, 4)$ $r = \sqrt{x^2 + y^2} = \sqrt{2^2 + 4^2} = \sqrt{20} = \underline{2\sqrt{5}}$

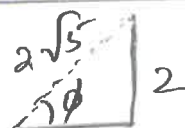
$$\tan \phi = \frac{y}{x} \Rightarrow \phi = \tan^{-1}\left(\frac{4}{2}\right) = \underline{\tan^{-1}(2)}$$

$$\hat{r} = \cos \phi \hat{i} + \sin \phi \hat{j}$$

$$\Rightarrow \hat{r} = \frac{1}{\sqrt{5}} \hat{i} + \frac{2}{\sqrt{5}} \hat{j} = \underline{\underline{\frac{1}{\sqrt{5}} (\hat{i} + 2\hat{j})}}$$

$$\hat{\phi} = -\sin \phi \hat{i} + \cos \phi \hat{j}$$

$$\hat{\phi} = \underline{\underline{\frac{1}{\sqrt{5}} (-2\hat{i} + \hat{j})}}$$



$$\tan \phi = \frac{4}{2}$$

$$\cos \phi = \frac{2}{2\sqrt{5}} = \frac{1}{\sqrt{5}}$$

$$\sin \phi = \frac{4}{2\sqrt{5}} = \frac{2}{\sqrt{5}}$$

(ii) $(x, y) = (3, 6)$

$$r = \sqrt{3^2 + 6^2} = \sqrt{45} = \underline{3\sqrt{5}}$$

$$\phi = \tan^{-1}\left(\frac{6}{3}\right) = \underline{\tan^{-1}(2)}$$

$$\hat{r} = \frac{1}{\sqrt{5}} \hat{i} + \frac{2}{\sqrt{5}} \hat{j} = \underline{\underline{\frac{1}{\sqrt{5}} (\hat{i} + 2\hat{j})}}$$

$$\hat{\phi} = \underline{\underline{\frac{1}{\sqrt{5}} (-2\hat{i} + \hat{j})}}$$

(iii) $(x, y) = (-4, -2)$

$$r = \sqrt{(-4)^2 + (-2)^2} = \sqrt{20} = 2\sqrt{5}$$

$$\phi = \tan^{-1}\left(\frac{-2}{-4}\right) = \tan^{-1}\left(\frac{1}{2}\right)$$

$\therefore (-4, -2)$ lies in 3rd quadrant in Cartesian

the $\phi_{\text{correct}} = \pi + \tan^{-1}\left(\frac{1}{2}\right)$.

$$\hat{r} = -\frac{1}{\sqrt{5}}(2\hat{i} + \hat{j})$$

$$\hat{\phi} = \frac{1}{\sqrt{5}}(\hat{i} - 2\hat{j})$$

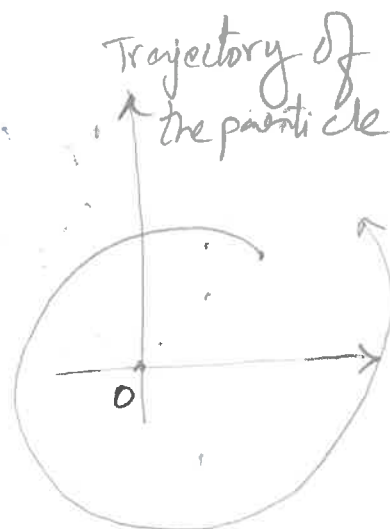
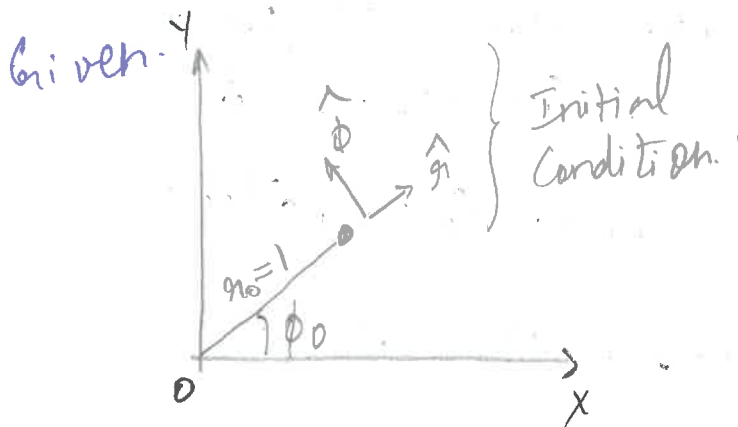
2) Given $v_r = 2t$ m/s & $\omega = 1$ rad/s. find

(i) $r(t)$ as a function of time if r at $t=0$ is 1m

(ii) $\phi(t)$ as a function of time if ϕ at $t=0$ is ϕ_0

(iii) At what distance from origin is the radial acceleration zero?

Sol



(i) $v_r = 2t \text{ m/s}$

$$\frac{dr}{dt} = 2t \text{ m/s}$$

$$dr = 2t dt$$

$$\int dr = \int 2t dt$$

Use initial
Condition
 $r(t=0) = 1 \text{ m}$

$$r(t) = \frac{2t^2}{2} + C$$

$$r(t) = t^2 + C \Rightarrow C = 1$$

$$\therefore (r(t) = t^2 + 1) \text{ m}$$

(ii) $\hat{\phi} = f(t)$

$$\hat{\phi}(t) = \underbrace{-\sin\phi \hat{i} + \cos\phi \hat{j}}_{\text{known}}$$

given $\omega = 1$

$$\omega = \frac{d\phi}{dt} = 1$$

$$\therefore \int d\phi = \int dt$$

$$\phi(t) = t + C \quad \text{given at } t=0 \quad \phi = \phi_0 \Rightarrow C = \phi_0$$

$$\therefore \phi(t) = t + \phi_0 \quad \text{or} \quad \phi_0 + t$$

$$\therefore \hat{\phi}(t) = \underline{-\sin(\phi_0 + t) \hat{i} + \cos(\phi_0 + t) \hat{j}}$$

(iii) a_r • radial acceleration is zero

$$\vec{a} = (a_r - r\omega^2) \hat{r} + (2v_r\omega + r\dot{\omega}) \hat{\phi}$$

↳ acceleration in polar Coordinates.

If radial acceleration is zero

$$\Rightarrow a_r - r\omega^2 = 0$$

we know $v_r = 2t$

$$\therefore a_r = \frac{dv_r}{dt} = \frac{d(2t)}{dt} = 2$$

we know $\omega = 1$

$\therefore$ the radial component is now

$$a_r - r\omega^2 = 2 - r \cdot (1)^2 = 0$$

$$\Rightarrow 2 - r = 0$$

$$r = \underline{2m}$$

$\therefore$ At $r = 2m$ from origin the radial acceleration is zero.